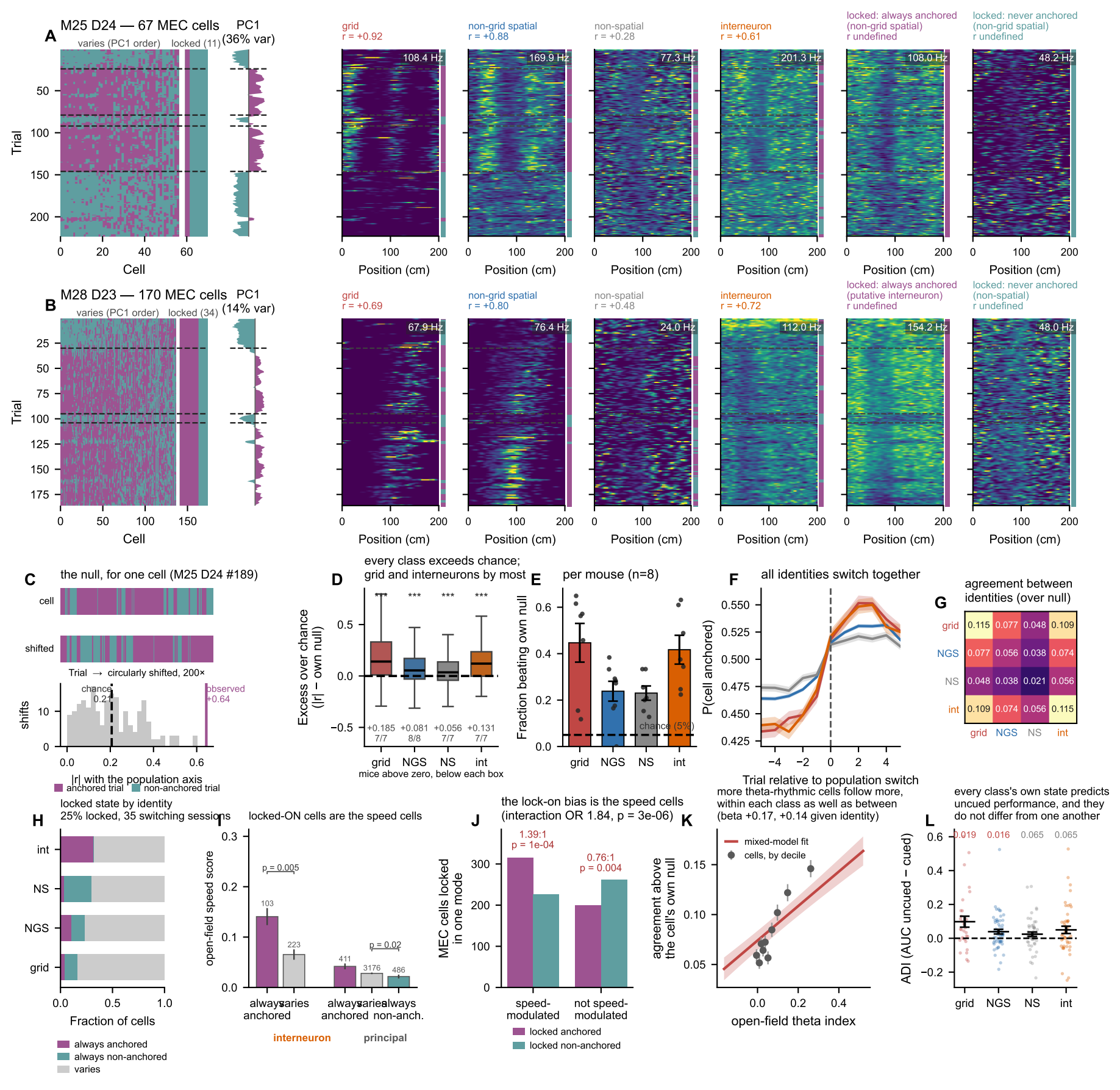

A
M25 D24 — 67 MEC cells
varies (PC1 order)
locked (11)
PC1 (36% var)
Trial
Cell
grid
r = +0.92
108.4 Hz
non-grid spatial
r = +0.88
169.9 Hz
non-spatial
r = +0.28
77.3 Hz
interneuron
r = +0.61
201.3 Hz
locked: always anchored (non-grid spatial)
r undefined
108.0 Hz
locked: never anchored (non-grid spatial)
r undefined
48.2 Hz
Position (cm)
B
M28 D23 — 170 MEC cells
varies (PC1 order)
locked (34)
PC1 (14% var)
grid
r = +0.69
67.9 Hz
non-grid spatial
r = +0.80
76.4 Hz
non-spatial
r = +0.48
24.0 Hz
interneuron
r = +0.72
112.0 Hz
locked: always anchored (putative interneuron)
r undefined
154.2 Hz
locked: never anchored (non-spatial)
r undefined
48.0 Hz
C
the null, for one cell (M25 D24 #189)
cell
shifted
Trial → circularly shifted, 200×
shifts
chance 0.21
observed +0.64
|r| with the population axis
anchored trial
non-anchored trial
D
every class exceeds chance;
grid and interneurons by most
Excess over chance (|r| − own null)
+0.185 7/7
+0.081 8/8
+0.056 7/7
+0.131 7/7
grid
NGS
NS
int
mice above zero, below each box
E
per mouse (n=8)
Fraction beating own null
chance (5%)
F
all identities switch together
P(cell anchored)
Trial relative to population switch
G
agreement between identities (over null)
grid 0.115 0.077 0.048 0.109
NGS 0.077 0.056 0.038 0.074
NS 0.048 0.038 0.021 0.056
int 0.109 0.074 0.056 0.115
H
locked state by identity
25% locked, 35 switching sessions
Fraction of cells
always anchored
always non-anchored
varies
I
locked-ON cells are the speed cells
open-field speed score
p = 0.005
103
223
p = 0.02
411
3176
486
always anchored
varies
always non-anch.
interneuron
principal
J
the lock-on bias is the speed cells
(interaction OR 1.84, p = 3e-06)
1.39:1
p = 1e-04
0.76:1
p = 0.004
MEC cells locked in one mode
speed-modulated
not speed-modulated
locked anchored
locked non-anchored
K
more theta-rhythmic cells follow more,
within each class as well as between
(beta +0.17, +0.14 given identity)
mixed-model fit
cells, by decile
agreement above the cell's own null
open-field theta index
L
every class's own state predicts
uncued performance, and they
do not differ from one another
0.019
0.016
0.065
0.065
ADI (AUC uncued − cued)